

Log_X — Chemical Reactor Yield Prediction

Predictive Modeling & Optimization in a Non-Isothermal Continuous Flow Reactor

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Training Dataset	150 Labeled Observations	Test Dataset	50 Hidden Observations
Target Variable	overall_yield (Product B %)	Selected Final Model	XGBoost Regressor
Best 5-Fold CV RMSE	20.1318 ($\pm$ 1.9719)	Submission Output	Log_X.csv (50 $\times$ 1)

1. Executive Summary

This project establishes a machine-learning surrogate model for a non-isothermal continuous-flow chemical reactor. The reactor operates a series-parallel reaction system where Reactant A converts to desired Product B, which can further undergo a side reaction to undesired Product C. The core objective is predicting the final yield percentage of Product B based on operating process variables.

The workflow seamlessly integrates chemical process engineering domain knowledge with exploratory data analysis, physics-inspired feature engineering, nonlinear machine learning algorithms, and rigorous cross-validation. The final selected model—an **XGBoost Regressor**—achieved a superior 5-fold cross-validation RMSE of **20.1318** (std: 1.9719), outperforming Random Forest, Decision Trees, Gradient Boosting, and Linear Regression.

2. Problem Definition and Chemical System

The reaction system is modeled as a series-parallel network within a non-isothermal continuous-flow reactor:



(Desired: $A \rightarrow B$ | Side Reaction: $B \rightarrow C$)

The challenge requires learning the complex, non-linear kinetics and thermodynamic trade-offs governing temperature distribution and residence time. Phase 1 focuses on quantitative prediction accuracy (RMSE across 50 hidden test samples), while Phase 2 mandates technical explanations of physical trade-offs, feature engineering rationale, robustness, and scalability.

3. Dataset Overview

The training dataset comprises 150 instances with 5 original process inputs and 1 target variable. Data integrity checks confirmed zero missing values and zero duplicate rows.

Variable Name	Role	Units	Engineering Meaning
flow_rate_L_min	Input	L/min	Volumetric flow rate of reactant mixture
concentration_mol_L	Input	mol/L	Inlet concentration of Reactant A
inlet_temperature_K	Input	K	Feed temperature entering the reactor
length_m	Input	m	Physical length of continuous reactor
jacket_temperature_K	Input	K	External heating/cooling jacket temperature
overall_yield	Target	%	Final yield percentage of Product B (Mean: 36.22, Min: 0.00, Max: 99.97)

4. Exploratory Data Analysis & Feature Engineering

Bivariate correlation analysis revealed significant negative linear correlations between thermal variables and overall yield, indicating that excessive heating promotes the undesirable side reaction $B \rightarrow C$.

Feature Name	Type	Correlation with overall_yield
jacket_temperature_K	Original Input	-0.4984
inlet_temperature_K	Original Input	-0.4051
Temp_Difference	Engineered	-0.1292
length_m	Original Input	+0.0800
Residence_Index	Engineered	+0.0764
Heat_Exposure	Engineered	-0.0551
flow_rate_L_min	Original Input	+0.0381
concentration_mol_L	Original Input	+0.0087

Physics-Informed Engineered Features

To capture chemical reactor physics without over-parameterization, three domain-specific variables were engineered:

Feature Name	Mathematical Formula	Physical Rationale & Purpose
Residence_Index	$Residence_Index = length_m / flow_rate_L_min$	Proxy for residence time and contact path opportunity within reactor.

Feature Name	Mathematical Formula	Physical Rationale & Purpose
Temp_Difference	$Temp_Difference = jacket_temperature_K - inlet_temperature_K$	Thermal driving force between utility jacket and feed stream.
Heat_Exposure	$Heat_Exposure = Temp_Difference \times Residence_Index$	Combined metric for total thermal energy impact during residence.

5. Model Benchmarking & Evaluation

Preliminary Holdout Results

Model / Experiment	RMSE	MAE	R ² Score
Linear Regression	31.4408	26.9214	0.4378
Decision Tree	30.6287	18.1755	0.4665
Random Forest	20.0320	13.1711	0.7718

5-Fold Cross-Validation Comparison

Due to the small sample size (150 rows), robust 5-fold cross-validation was conducted across ensemble algorithms:

Model	Fold 1	Fold 2	Fold 3	Fold 4	Fold 5	Mean RMSE	Std Dev
Random Forest	20.2082	25.8079	22.0932	18.4306	17.4188	20.7917	2.9715
Gradient Boosting	23.4087	24.1407	23.4752	20.8589	16.9703	21.7708	2.6488
XGBoost (Selected)	21.4936	20.4471	22.8786	18.1992	17.6403	20.1318	1.9719

6. Hyperparameter Tuning & Strategy

GridSearchCV Finding: Avoiding Overfitting

A GridSearchCV experiment was executed searching parameters: `n_estimators: 200`, `learning_rate: 0.05`, `max_depth: 4`, `subsample: 0.8`, `colsample_bytree: 1.0`. This produced a CV RMSE of **22.4175**, which performed worse than default XGBoost (**20.1318**). Consequently, the default setup was preserved to prevent hyperparameter overfitting on the small training dataset.

Final Selected Architecture

`XGBRegressor(n_estimators=100, learning_rate=0.1, max_depth=4, random_state=42)`

7. Engineering Interpretation & Project Limitations

Chemical Dynamics: Feature importance analysis ranked `inlet_temperature_K` (0.3730), `jacket_temperature_K` (0.3188), and `Residence_Index` (0.1487) as the top 3 drivers of yield. This aligns with kinetics: elevated thermal exposure accelerates the formation of B, but excessive temperatures or prolonged residence times push the equilibrium toward the degradation side-product C.

Project Limitations:

- Small sample size (150 rows) introduces statistical variance in validation estimates.
- Test labels (50 rows) remain hidden; leaderboard score depends on generalization resilience.
- Feature proxies describe empirical behavior but do not replace full first-principles reaction kinetics.

8. Conclusion & Reproducibility Summary

The project successfully deployed a physics-aware machine learning workflow for continuous reactor optimization, achieving a winning CV RMSE of **20.1318**. All submission constraints (50 × 1 output in `Log_X.csv`) were validated.

Component / Item	Recorded Value / Parameter
Notebook Reference	<code>Log_X.ipynb</code>
Training / Test Observations	150 / 50
Engineered Features	<code>Residence_Index</code> , <code>Temp_Difference</code> , <code>Heat_Exposure</code>
Selected Algorithm	<code>XGBRegressor</code> (<code>n_estimators=100</code> , <code>learning_rate=0.1</code> , <code>max_depth=4</code>)
Best Cross-Validation RMSE	20.131763 (Std Dev: 1.971873)
Final Output Verification	<code>Log_X.csv</code> , Shape (50, 1), 100% Continuous Numerical Format

Source basis: Data and details synthesized from `Log_X.ipynb` notebook and official Chemical Reactor Yield Prediction Hackathon specification.